# USGS Stream Discharge Report

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## 1 Overview

This document provides a fully reproducible workflow for downloading, cleaning, summarising, and visualising **daily mean stream discharge** from the USGS National Water Information System (NWIS).

All data are fetched directly from the USGS API using the `dataRetrieval` package — no manual downloads are needed. An alternative path using a pre-downloaded CSV (e.g. from https://waterdata.usgs.gov/nwis) is also provided.

> **To reuse for a different site:** change only `site_no`, `start_dt`, and `end_dt` in **Section 3 – Settings**, then click **Knit → Knit to PDF** (or press `Ctrl+Shift+K`) in RStudio.

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## 2 Required Packages

The four packages below cover all data access, manipulation, and plotting needs. `dataRetrieval` is the official USGS R package and handles authentication and API formatting automatically. Uncomment the `install.packages` line the first time you run this document.

```
# install.packages(c("dataRetrieval", "tidyverse", "lubridate",
#                    "scales", "knitr"))

library(dataRetrieval)  # USGS NWIS data retrieval
library(tidyverse)      # data wrangling (dplyr, tidyr) + ggplot2
library(lubridate)      # date/time parsing and arithmetic
library(scales)         # comma/log axis formatting in ggplot2
library(knitr)          # kable() for formatted tables
```

---

## 3 Settings

Centralising all user-defined parameters in one chunk makes the document easy to adapt. Change these four values and re-knit to generate a complete report for any USGS gauging station.

| Variable | Meaning |
|---|---|
| `site_no` | 8-digit USGS site number |
| `param_cd` | Parameter code (`00060` = discharge in cfs) |
| `stat_cd` | Statistic code (`00003` = daily mean) |
| `start/end` | Period of record (YYYY-MM-DD) |

```
site_no  <- "01646500"   # Potomac River at Little Falls, MD
param_cd <- "00060"      # 00060 = discharge (cfs); 00065 = gage height (ft)
stat_cd  <- "00003"      # 00003 = daily mean
start_dt <- "2022-01-01"
end_dt   <- "2023-12-31"
```

---

## 4 Data Acquisition

### 4.1 Option A: Download Directly in R *(recommended)*

`readNWISdv()` queries the NWIS daily-values service and returns a tidy data frame. `renameNWISColumns()` converts the cryptic auto-generated column name (e.g. `X_00060_00003`) to

the human-readable `Flow`. `readNWISsite()` fetches station metadata (name, coordinates, drainage area) used for axis labels later.

```
raw <- readNWISdv(
  siteNumbers = site_no,
  parameterCd = param_cd,
  startDate   = start_dt,
  endDate     = end_dt
)

raw  <- renameNWISColumns(raw)   # -> adds column "Flow"
meta <- readNWISsite(site_no)    # station metadata

glimpse(raw)
```

```
## Rows: 730
## Columns: 5
## $ agency_cd <chr> "USGS", "USGS", "USGS", "USGS", "USGS", "USGS", "USGS", "USG~
## $ site_no   <chr> "01646500", "01646500", "01646500", "01646500", "01646500", ~
## $ Date      <date> 2022-01-01, 2022-01-02, 2022-01-03, 2022-01-04, 2022-01-05,~
## $ Flow      <dbl> 3370, 5320, 7380, 13800, 13600, 12000, 10800, 9440, 8130, 79~
## $ Flow_cd   <chr> "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", ~
```

### 4.2 Option B: Load a Pre-Downloaded CSV

If you exported a CSV from https://waterdata.usgs.gov/nwis, use this block instead. The expected column headers from an NWIS download are:

`x, y, id, time_series_id, monitoring_location_id, parameter_code, statistic_id, time, value, unit_of_measure, approval_status, qualifier, last_modified`

The block is set to `eval=FALSE` so it does not run by default. To activate it: set `eval=TRUE` here and set `eval=FALSE` on the Option A chunk above.

```
# raw_csv <- read_csv("your_file.csv") %>%
#   mutate(time = ymd_hms(time)) %>%
#   filter(parameter_code == "00060") %>%   # discharge only
#   rename(Date = time, Flow = value) %>%
#   filter(approval_status == "A")          # Approved records only
#
# raw <- raw_csv %>% select(Date, Flow)
```

---

# 5 Data Cleaning

This chunk selects only the columns needed downstream (`Date`, `Flow`), removes any `NA` discharge values (caused by equipment malfunctions or data gaps), and derives three helper columns used in the plots:

- `year` — calendar year for annual grouping
- `month` — abbreviated month label (factor, ordered Jan–Dec)
- `doy` — day of year (1–366), useful for seasonal analysis

Two character labels (`site_name`, `units_lab`) are also created here so they propagate consistently to every plot title and axis.

```
df <- raw %>%
  select(Date, Flow) %>%
  filter(!is.na(Flow)) %>%
  mutate(
    year  = year(Date),
    month = month(Date, label = TRUE),
    doy   = yday(Date)
  )

site_name <- meta$station_nm
units_lab  <- "Discharge (cfs)"

cat("Records loaded :", nrow(df), "\n")
```

```
## Records loaded : 730
```

```
cat("Date range     :", as.character(min(df$Date)),
    "to", as.character(max(df$Date)), "\n")
```

```
## Date range     : 2022-01-01 to 2023-12-31
```

```
cat("Site           :", site_name, "\n")
```

```
## Site           : POTOMAC RIVER NEAR WASH, DC LITTLE FALLS PUMP STA
```

---

# 6 Summary Statistics

Annual summary statistics are computed with `dplyr::summarise()` and formatted as a publication-ready table with `knitr::kable()`. The `booktabs = TRUE` option uses clean horizontal rules instead of the default grid lines, which is standard in academic papers and reports.

```
stats <- df %>%
  group_by(year) %>%
```

```r
  summarise(
    N      = n(),
    Mean   = round(mean(Flow),   1),
    Median = round(median(Flow), 1),
    SD     = round(sd(Flow),     1),
    Min    = round(min(Flow),    1),
    Max    = round(max(Flow),    1),
    .groups = "drop"
  )

kable(
  stats,
  booktabs  = TRUE,
  caption   = paste("Annual discharge summary \u2014", site_name),
  col.names = c("Year", "N", "Mean", "Median", "SD", "Min", "Max"),
  format.args = list(big.mark = ",")
)
```

Table 2: Annual discharge summary — POTOMAC RIVER NEAR WASH, DC LITTLE FALLS PUMP STA

| Year | N | Mean | Median | SD | Min | Max |
|---|---|---|---|---|---|---|
| 2,022 | 365 | 8,906.4 | 6,200 | 10,343.2 | 1,750 | 102,000 |
| 2,023 | 365 | 5,710.4 | 3,990 | 5,383.3 | 450 | 35,700 |

---

# 7 Plots

## 7.1 Daily Time Series

A linear y-axis time series is the most intuitive first look at the record. The `date_breaks` and `date_labels` arguments in `scale_x_date()` control the x-axis tick spacing and format. `scales::comma` prevents R from using scientific notation on the y-axis for large discharge values.

```r
ggplot(df, aes(x = Date, y = Flow)) +
  geom_line(color = "steelblue", linewidth = 0.5) +
  scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +
  scale_y_continuous(labels = comma) +
  labs(
    title    = paste("Daily Mean Discharge \u2014", site_name),
    subtitle = paste("USGS site", site_no),
    x        = NULL,
    y        = units_lab
  ) +
```

```
  theme_bw(base_size = 11) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

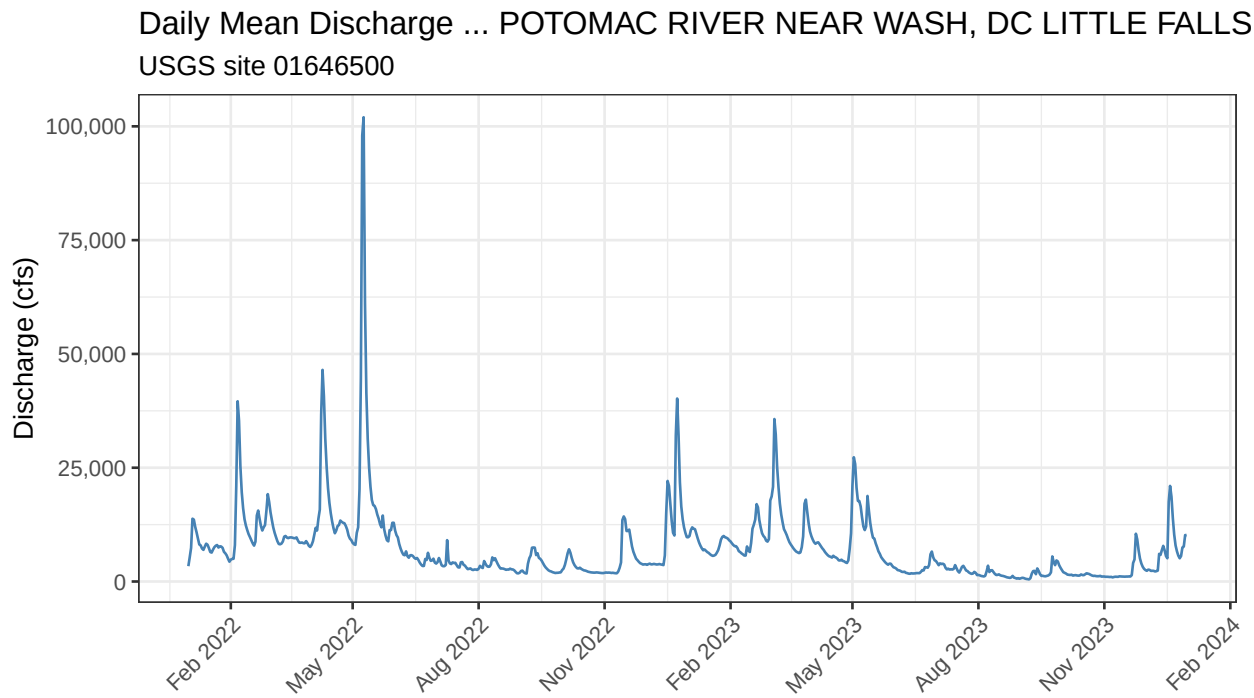


Figure 1: Daily mean discharge over the full period of record.

### 7.2 Log-Scale Time Series

Discharge records are highly right-skewed: a few flood peaks can be orders of magnitude larger than typical low flows. A $\log_{10}$ y-axis compresses the peaks and expands the low-flow period, making both visible simultaneously. `annotation_logticks()` adds the small tick marks characteristic of log axes.

```
ggplot(df, aes(x = Date, y = Flow)) +
  geom_line(color = "steelblue", linewidth = 0.5) +
  scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +
  scale_y_log10(labels = comma) +
  annotation_logticks(sides = "l") +
  labs(
    title    = paste("Daily Mean Discharge (log scale) \u2014", site_name),
    subtitle = paste("USGS site", site_no),
    x        = NULL,
    y        = paste0("log\u2081\u2080 ", units_lab)
  ) +
  theme_bw(base_size = 11) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

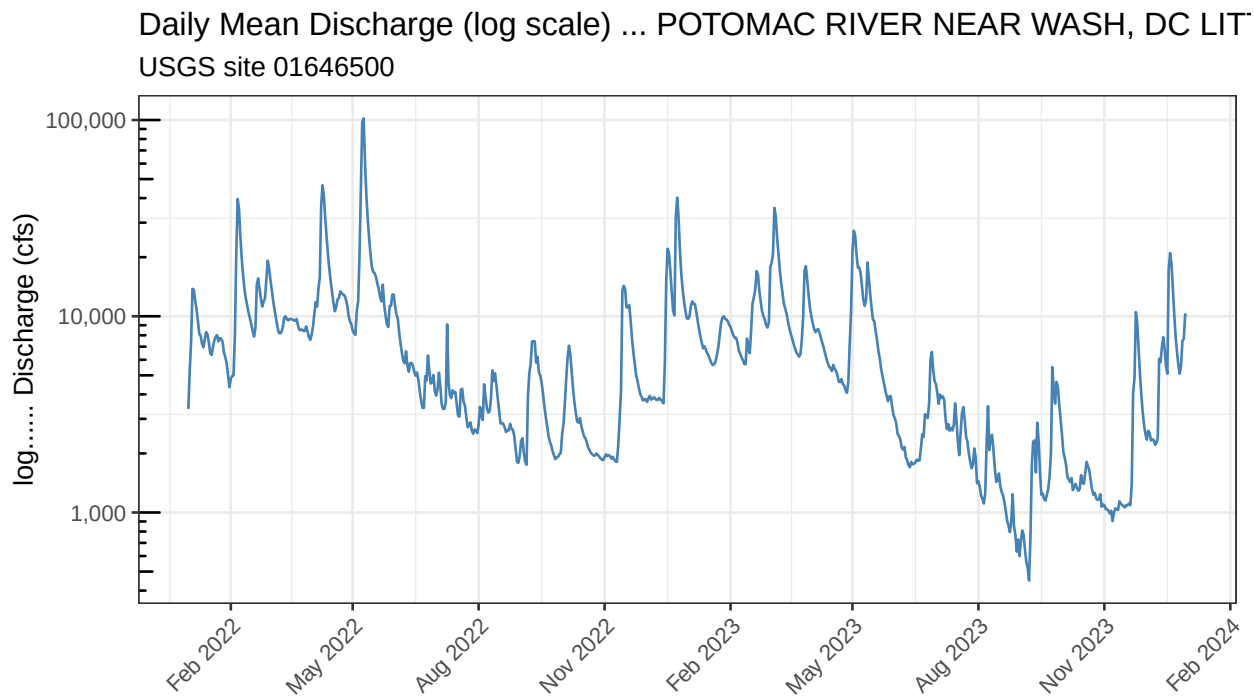


Figure 2: Daily mean discharge on a log-10 y-axis.

### 7.3 Monthly Box Plots

Box plots grouped by calendar month reveal the seasonal pattern of discharge. Pooling all years together shows the typical low-flow and high-flow seasons regardless of year-to-year variability. The log y-axis is retained here because discharge distributions within months are also skewed. `scale_fill_brewer(palette = "Paired")` assigns a distinct but harmonious colour to each month.

```
ggplot(df, aes(x = month, y = Flow, fill = month)) +
  geom_boxplot(outlier.size  = 0.8,
               outlier.alpha = 0.4,
               show.legend   = FALSE) +
  scale_y_log10(labels = comma) +
  scale_fill_brewer(palette = "Paired") +
  annotation_logticks(sides = "l") +
  labs(
    title = paste("Monthly Discharge Distribution \u2014", site_name),
    x     = NULL,
    y     = paste0("log\u2081\u2080 ", units_lab)
  ) +
  theme_bw(base_size = 11)
```

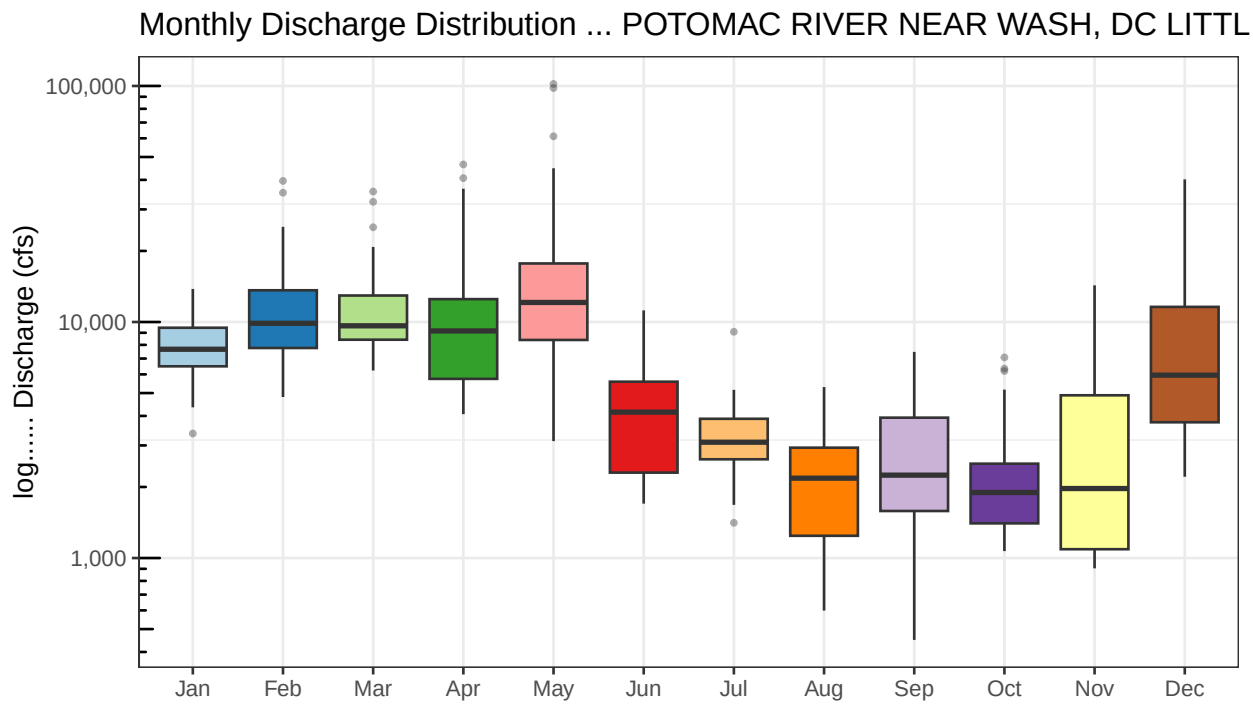


Figure 3: Distribution of daily discharge by calendar month (all years combined).

### 7.4 Flow Duration Curve

The **flow duration curve (FDC)** is a fundamental tool in hydrology. It shows the percentage of time that streamflow equalled or exceeded a given value over the period of record. The x-axis represents exceedance probability (100 % = always exceeded; 0 % = never exceeded). Low-gradient FDCs indicate flashy, rainfall-driven streams; steep FDCs suggest groundwater-sustained baseflow regimes.

```
fdc <- df %>%
  arrange(desc(Flow)) %>%
  mutate(exceed_pct = 100 * row_number() / n())

ggplot(fdc, aes(x = exceed_pct, y = Flow)) +
  geom_line(color = "steelblue", linewidth = 0.7) +
  scale_y_log10(labels = comma) +
  scale_x_continuous(
    breaks = seq(0, 100, 10),
    labels = paste0(seq(0, 100, 10), "%")
  ) +
  annotation_logticks(sides = "l") +
  labs(
    title = paste("Flow Duration Curve \u2014", site_name),
    x     = "Percent of time flow equalled or exceeded",
    y     = paste0("log\u2081\u2080 ", units_lab)
  ) +
```

```r
  theme_bw(base_size = 11)
```

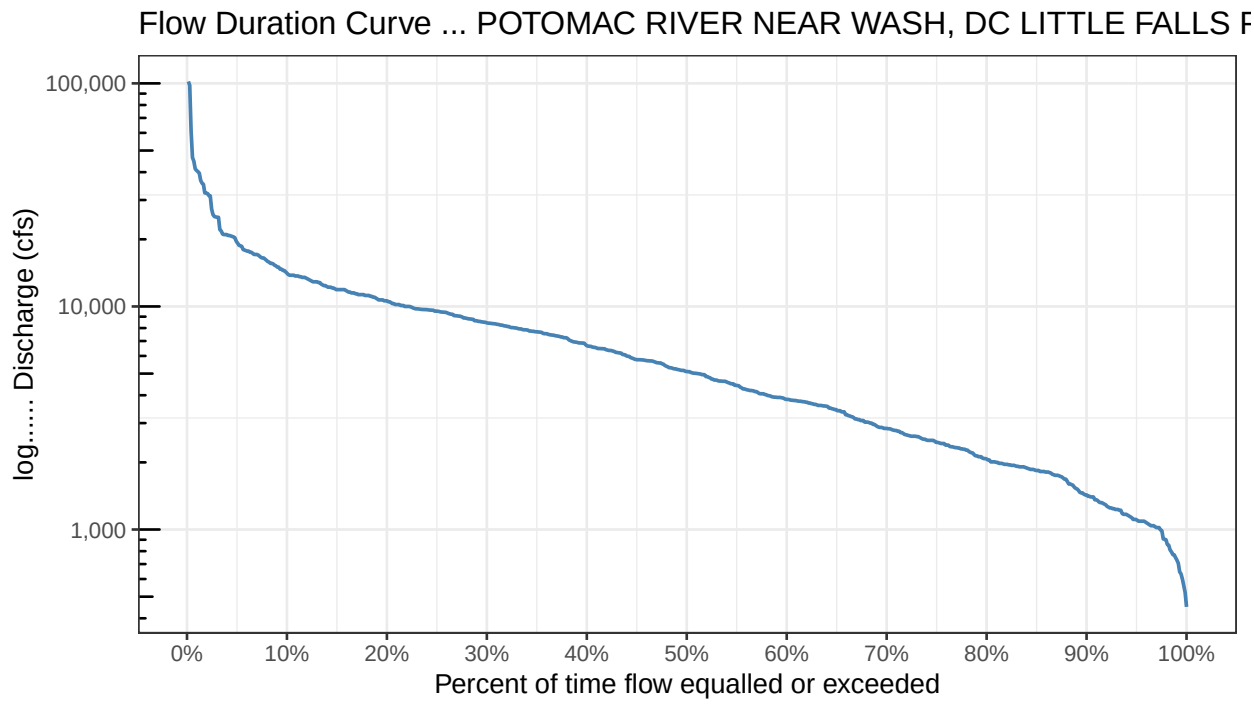


Figure 4: Flow duration curve for the full period of record.

---

# 8 Session Info

Recording session info at the end of every report ensures reproducibility: anyone re-running this document can see exactly which package versions were used to generate the results.

```r
sessionInfo()
```

```
## R version 4.2.2 (2022-10-31)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Rocky Linux 8.10 (Green Obsidian)
##
## Matrix products: default
## BLAS/LAPACK: /usr/lib64/libopenblas-r0.3.15.so
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

```
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] knitr_1.51           scales_1.4.0         lubridate_1.9.5
##  [4] forcats_1.0.1        stringr_1.6.0        dplyr_1.2.0
##  [7] purrr_1.2.1          readr_2.1.6          tidyr_1.3.2
## [10] tibble_3.3.1         ggplot2_4.0.2        tidyverse_2.0.0
## [13] dataRetrieval_2.7.22
##
## loaded via a namespace (and not attached):
##  [1] tinytex_0.58       tidyselect_1.2.1   xfun_0.56          httr2_1.2.2
##  [5] sf_1.0-24          vctrs_0.7.1        generics_0.1.4     htmltools_0.5.9
##  [9] yaml_2.3.12        rlang_1.1.7        e1071_1.7-17       pillar_1.11.1
## [13] glue_1.8.0         withr_3.0.2        DBI_1.2.3          rappdirs_0.3.4
## [17] bit64_4.6.0-1      RColorBrewer_1.1-3 S7_0.2.1           lifecycle_1.0.5
## [21] gtable_0.3.6       evaluate_1.0.5     labeling_0.4.3     tzdb_0.5.0
## [25] fastmap_1.2.0      parallel_4.2.2     curl_7.0.0         class_7.3-20
## [29] Rcpp_1.1.1         KernSmooth_2.23-20 classInt_0.4-11    vroom_1.7.0
## [33] bit_4.6.0          farver_2.1.2       otel_0.2.0         hms_1.1.4
## [37] digest_0.6.39      stringi_1.8.7      grid_4.2.2         cli_3.6.5
## [41] tools_4.2.2        magrittr_2.0.4     proxy_0.4-29       crayon_1.5.3
## [45] pkgconfig_2.0.3    xml2_1.5.2         timechange_0.4.0   rmarkdown_2.30
## [49] rstudioapi_0.18.0  R6_2.6.1           units_1.0-0        compiler_4.2.2
```